# FIReVision:

## A Photorealistic Thermal IR Simulator for Wildfire Robotics

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PORGES FELLOW
*This project was made possible by the generosity of the* **Porges Family Fund for Undergraduate Research**

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# Wildfires Are A Growing Challenge



Wildfires are becoming more frequent and complex



Improved situational awareness is crucial for firefighters

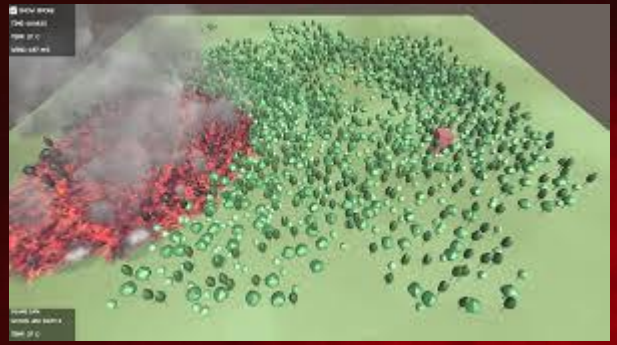


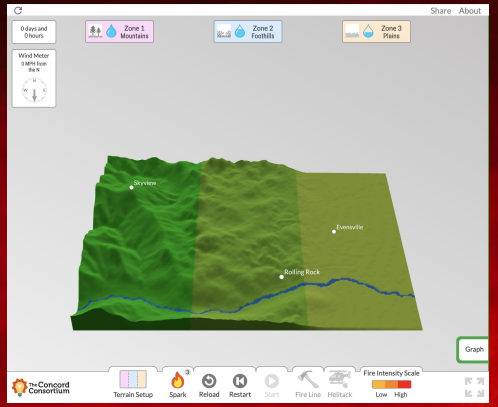
Unmanned aerial systems (UAS) offer a good solution for monitoring wildfire environments.

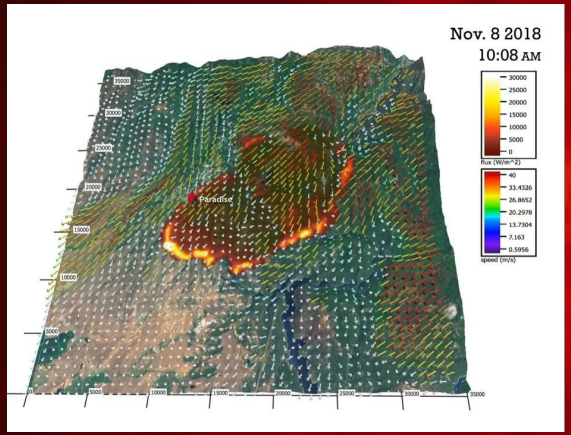
# There is a heavy need advanced Wildfire Simulators

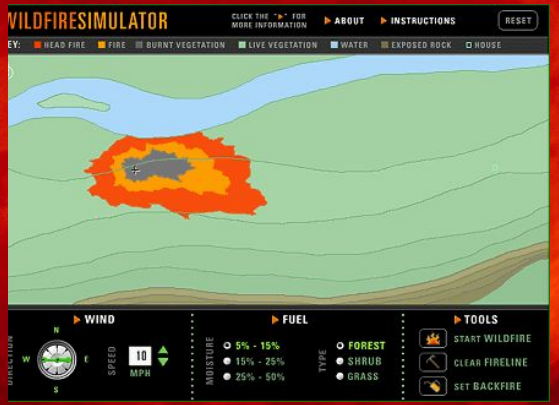
Very few exist for UAS.
The ones that do exist:

- Are limited in capabilities and realism.
- Restricted to 2D space.
- Lack realistic UAS dynamics.
- Do not include important targets like crew and vehicles
- Usually lack a camera-based perception system.





# FIReVision is a photorealistic thermal infrared simulator.

FIReVision

AirSim



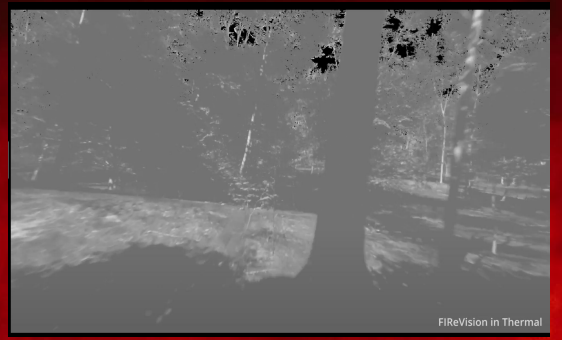

Unreal Engine 5

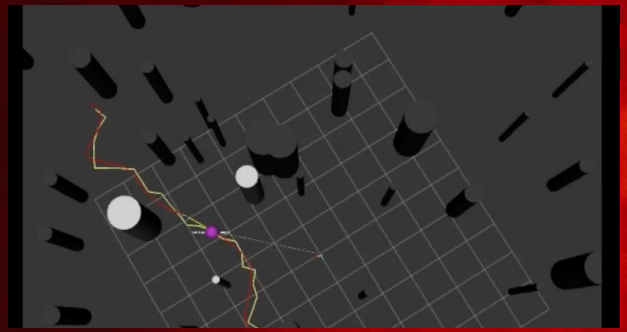


# FIReVision enables researchers to explore  advanced robotics tasks.

- Provides realistic RGB and Thermal Vision.
- Provides visually realistic, dynamic smoke.
- Poses challenges by being visually degrading, hazardous, and dynamic.
- Has Automatic Data Labeling with Object Detection & Depth Estimation.
- Is Open-Source & Accessible,





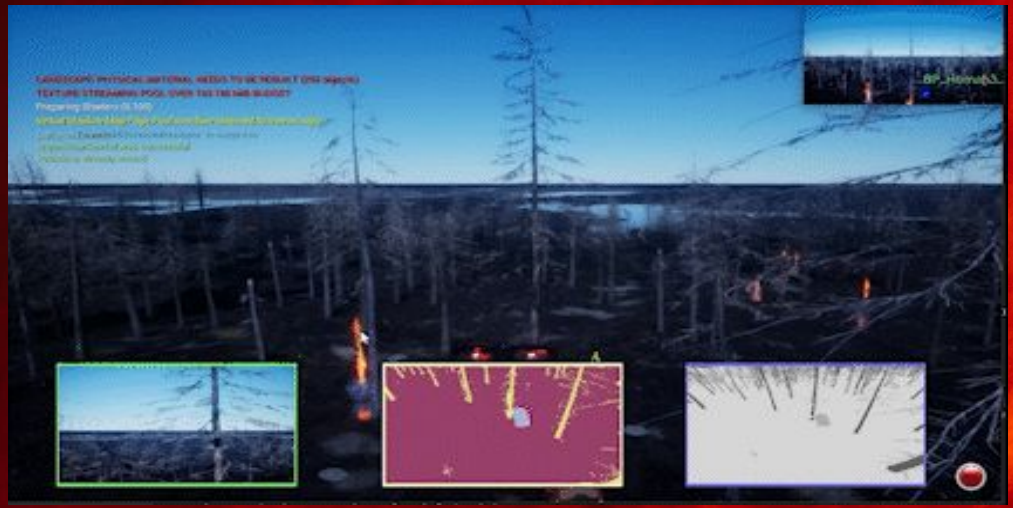





# Personal Contributions to FIReVision

Created Python Scripts that provided:

- Keyboard control system for simulated drone navigation.
- Real-time object detection for enhanced situational awareness
- Efficient data collection to generate diverse image datasets for object detection



# FIReVision Demo Video



# Thank You For Listening!

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